

The Time-Winding as a Logarithmic Spiral

The K_{frak} recursion forces self-similarity ratio e and continuous scale invariance —
and what HLV's spiral time would have to satisfy

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.1 Question: what exactly is FFGFT's time-winding?

In FFGFT, time is the mass circle, woven into the remaining circles via $\tilde{T} \cdot m = 1$ and not a separable factor (Dok. 285). A precise question follows: what exactly *is* FFGFT's time-winding?

This document settles it completely FFGFT-internally — it is *derivable* from the fractal correction. The result in one sentence: the K_{frak} recursion forces the time-winding into a *logarithmic spiral with self-similarity ratio e* , that is, into continuous scale invariance.

.2 Three building blocks from the corpus

Three quantities are fixed elsewhere and are only brought together here. First, the fractal dimension of space, $D_f = 3 - \xi_0$ with $\xi_0 = 4/30000$ (Dok. 205, Dok. 101). Second, the fractal correction in the gate: the X rotation is not by π but by $\alpha = \pi K_{\text{frak}}$ with $K_{\text{frak}} = 1 - 100 \xi_0$ (Dok. 034). Third, the scale run as the recursion $\xi_{n+1} = \xi_n (1 - 100 \xi_0)$, more precisely $\xi_{n+1} = \xi_n (1 - 100 \xi_n)$ (Dok. 146).

From $\xi_0 = 1/7500$ follow three exact facts that carry the rest:

$$100 \xi_0 = \frac{1}{75}, \quad K_{\text{frak}} = \frac{74}{75}, \quad \frac{1}{100 \xi_0} = 75. \quad (1)$$

.3 Non-closure: from the 75-closure to the spiral

A winding that should advance by 2π per turn advances gate-wise only by πK_{frak} and *misses* π by $100\pi \xi_0$ per half-turn. Iterated, the winding does not close but precesses — it becomes a spiral. The closure defect per turn is $d = 100 \xi_0$ in turn units.

For frozen ξ_0 , $d = 1/75$ is constant, so the rotation number $74/75$ is rational: the winding closes after *exactly 75 turns* and carries a 75-fold structure — the number $75 = 1/(100 \xi_0)$ drops out exactly. The limiting case of constant pitch, $d = 0$, closes already after one turn. Both closed cases are the *rolled-up*, compact reading of the time-winding: a single scale, the winding lies closed and is static. FFGFT already knows a non-closure algebraically, in the non-multiplicative mode composition (Dok. 193); here its geometric-temporal counterpart is added.

.4 The scale run makes the spiral logarithmic

When ξ runs over the recursion, the rational 75-closure turns into a non-closing, self-similar spiral. This transition is the *unrolling* of the winding: the spiral is the decompactified counterpart of the closed 75-winding, and "unrolling" here means precisely letting the scale run (the recursion), not merely splitting off a torus factor. The derivation is closed. With $u_n = 1/\xi_n$,

$$u_{n+1} = \frac{1}{\xi_n(1 - 100 \xi_n)} = u_n (1 + 100 \xi_n + \mathcal{O}(\xi_n^2)) = u_n + 100 + \mathcal{O}(\xi_n), \quad (2)$$

so $u_n \approx 100n + 7500$ and hence an *algebraic* $1/n$ decay

$$\xi_n \approx \frac{1}{100(n + 75)}. \quad (3)$$

The cumulative precession defect thereby becomes logarithmic,

$$D(k) = \sum_{n \leq k} 100 \xi_n \approx \sum_{n \leq k} \frac{1}{n + 75} \approx \ln\left(1 + \frac{k}{75}\right). \quad (4)$$

With radius $\rho = k + 75$ — the pole of the spiral sits at the effective origin $n = -75 = -1/(100\xi_0)$, the same 75 as before — and precession angle $\beta = 2\pi D(k)$, it follows that $\rho/75 = e^{\beta/2\pi}$, that is

$$\rho = 75 e^{\beta/2\pi} = \text{logarithmic (equiangular) spiral.} \quad (5)$$

Its self-similarity is thereby fixed: per precession round, $\Delta\beta = 2\pi$, the radius scales by $e^1 = e$. The *ratio* e is not assumed but follows from the algebraic $1/n$ decay of the recursion: it is the base of the natural logarithm produced by the harmonic sum, and no additional parameter enters. The radius offset 75 is likewise not freely chosen — it is $1/(100\xi_0)$, the same constant that already carries the 75-closure.

Both are confirmed numerically without leeway (`zeit_logspirale_probe.py`): the fit $\ln \rho = a\beta + b$ gives $a = 0.15922 \approx 1/2\pi = 0.15915$ and $e^{2\pi a} = 2.7195 \approx e$; the locally measured self-similarity ratio converges scale by scale $2.734 \rightarrow 2.726 \rightarrow 2.721 \rightarrow 2.720 \rightarrow 2.719 \rightarrow 2.719$ towards $e = 2.71828$.

.5 Continuous, not discrete scale invariance

The self-similarity is *continuous* — a log spiral, a power law — not discretely log-periodic. A test for discrete scale invariance (the residual of $D(k)$ against the smooth $\ln(1 + k/75)$) yields an amplitude near zero and no periodicity in $\ln k$. This agrees with the corpus: Dok. 146 records that the scaling over the factor $1/\xi_0 = 7500$ becomes quasi-continuous through about 13 logarithmic sub-levels, and the fractal dimension stays constant without steps. The time-winding realises the same continuous scale invariance geometrically, as a log spiral.

.6 Space and time: the same ξ_0 , scaled differently

Space and time carry the same fundamental constant ξ_0 at two places: the spatial box-dimension deficit is ξ_0 (in $D_f = 3 - \xi_0$), the winding defect per gate is $100\xi_0 = 1/75$. The factor 100 between them is the recursion coefficient; it links a *geometric* deficit to a *dynamical* defect and privileges no axis. Which axis carries the defect is a matter of projection — it is conventionally noted on time, but is symmetric under $\tilde{T} \cdot m = 1$ (following sections), hence no amplification of time over space.

A boundary must be drawn explicitly here: the box dimension of the time-winding itself is ≈ 1 — it is a curve — and *not* $3 - \xi_0$. The link between the temporal spiral and the spatial fractal dimension is the shared ξ_0 , not an equality of dimensions. The time spiral does not carry the space dimension.

.7 The choice of axis is a projection

The logarithmic spiral is a property of the whole coupled winding on T^4 , not of a single coordinate. What is forced is the *non-closure* with ratio e ; onto which axis one places it is a matter of representation. Because $\tilde{T} \cdot m = 1$ couples time and mass inversely and the torus weaves all circles together — nothing factorises — there is no distinguished time dimension that rolls up on its own. A generic winding runs through several circles at once; the pure *time* winding is the special case of a projection. The same non-closure could equally be placed on a space axis or on a mixture — different basis, same invariant.

The assignment to time is thus an artificial shift onto one axis — a convention, not a distinction. The gate defect is conventionally described on the time-winding (Dok. 034);

the dual projection (next section) shows, however, that the same non-closure appears with the same factor 100 on the space side. No basis is thereby preferred; each is an equivalent representation of the same invariant — the same representational freedom as between the two equivalent representations (Dok. 206) and in the translation invariance of $\theta = 2/9$ (Dok. 293).

.8 Continuous time: the dual projection into space

If, conversely, time is set *continuous* — smooth, without gate-wise rolling up — then the non-closure can no longer sit on time; it must be carried by another axis. The coupling $\tilde{T} \cdot m = 1$ does not allow time to be smoothed in isolation: with $d\tilde{T}/\tilde{T} = -dm/m$ the mass runs along inversely, and with it the spatial-geometric side ($D_f = 3 - \xi_0$). The running winding then appears as a *variation of space* rather than as rolled-up time. It is the same invariant non-closure with ratio e , only rotated into the dual projection: in the present reading time runs (rolls up) and space stays comparable; in the dual reading time is continuous and space varies.

This mirroring has been worked out numerically (`dual_projektion_probe.py`). Because time advances per gate shortened by the factor $K_{\text{frak}} = 1 - 100\xi_n$, mass advances by the inverse factor $1/K_{\text{frak}}$ under $\tilde{T} \cdot m = 1$; the defect per step is $100\xi_n$ on both sides (the ratio $d_{\text{mass}}/d_{\text{time}} \rightarrow 1$ to six digits). The factor 100 therefore does not *move*, it is *mirrored*: the gate factors are inverse ($K_{\text{frak}} \leftrightarrow 1/K_{\text{frak}}$), the amplification 100 is the same. Both projections yield the same logarithmic growth, and the spiral fit gives the same ratio e for both ($e^{2\pi a} = 2.7195$ and 2.7192); their cumulative defects differ only by a *bounded* constant (≈ 0.0134 , not growing with $\ln k$). The core content is thereby representation-invariant and the non-closure preserved: time cannot be smoothed without mass and space taking over the winding — the mass-time duality principle in action. Its distribution over time, mass and space is a matter of projection; the ratio e is not.

.9 Placement relative to HLV

The FFGFT-internal statement is thereby *derived*: the K_{frak} recursion forces the time-winding to be a log spiral with ratio e . Only here does HLV enter. Time appears differently in the two frameworks (Dok. 285): HLV factorises $T^6 \times S^1$, where time is an A_5 singlet and decoupled as *spiral time*, a separable object; FFGFT does not factorise, its spiral is the projection of an inseparable whole. The limiting case of constant pitch ($d = 0$) noted above corresponds to the fixed point $\xi \rightarrow 0$ of the recursion, in which the defect vanishes and the winding closes — that is HLV's decoupled singlet time; the running FFGFT winding approaches it asymptotically.

Whether HLV's spiral time is the *same* object remains a candidate — but now sharply testable: it would have to be a log spiral with self-similarity ratio e , equivalent to a $1/n$ decay of its pitch. The decisive question to HLV's formulation (sources on spiral time) is therefore no longer "similar?" but "ratio e , yes or no?". Ordered within the three methodological layers: the derivation recursion $\rightarrow$ log spiral is a core derivation, obtained from ξ ; the identification with HLV's spiral time is a bridge candidate; no physical claim over HLV is made.

The contrast here is structural, not ontological. Both frameworks — and likewise the two variants within FFGFT, whether time or space is rolled up — are mathematical models, not reality; no factorisation and no choice of axis is thereby ontologically privileged. That one hopes such a model is applicable to reality is the motivation, not a statement about what is.

.10 Status

Secured and derived: the time-winding as a logarithmic spiral with self-similarity ratio e , continuous scale invariance, from the K_{frak} recursion; the exact 75-closure in the frozen case; the shared constant ξ_0 , appearing as the spatial box deficit ξ_0 and as the winding defect $100\xi_0$ (factor 100 = recursion coefficient, axis-symmetric). The closed 75-winding is here the rolled-up (compact, single-scale) reading, the log spiral the unrolled (decompactified, across-scales) reading of the same time-winding. What remains open is only whether HLV's spiral time carries the ratio e ; that decides the identification and is to be checked against HLV's spiral-time definition. Reproducible in `zeit_windung_probe.py`, `zeit_logspirale_probe.py` and `dual_projektion_probe.py`.